

Active Directory Lab – Part 2: Active Directory Domain Controller Setup

Objective

The objective of this lab was to configure the Windows Server virtual machine as an Active Directory domain controller. This included installing the required server roles, creating a new domain, and verifying that Active Directory could be managed successfully.

Lab Environment

This part of the lab used the Windows Server virtual machine configured in Part 1. The server was prepared with a static host-only IP address of 192.168.56.10 so that it could provide stable domain and DNS services to other lab machines.

The domain created for this lab was “adlab.local”, with the NetBIOS name “ADLAB”.

Tools Used:

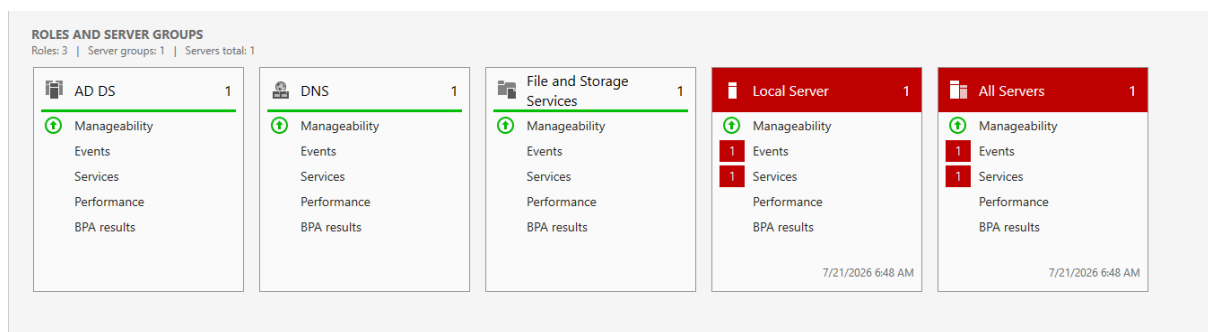
- VirtualBox
- Windows Server 2022 Evaluation
- Server Manager
- Active Directory Users and Computers
- Command Prompt

Steps Performed

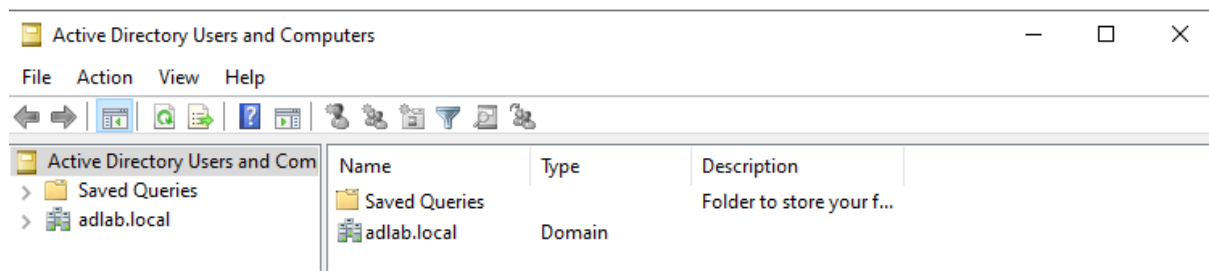
Active Directory Domain Services was installed through Server Manager using the Add Roles and Features wizard. During the installation, the required management tools were also added so the domain could be managed from the server.

The server was then promoted to a domain controller. During this process, a new forest was created and the server was configured to provide domain services for the lab environment. DNS was also configured as part of the domain controller setup so that domain clients could later locate and join the domain.

After the promotion process completed, the server restarted and loaded as a domain controller. Server Manager then showed the AD DS and DNS roles installed on the server.



Active Directory Users and Computers was opened to confirm that the domain had been created successfully. The adlab.local domain was visible, showing that Active Directory was ready for managing users, groups, computers, and organisational units.



Findings

The Windows Server virtual machine was successfully promoted to a domain controller. Server Manager confirmed that the AD DS and DNS roles were installed, and Active Directory Users and Computers confirmed that the adlab.local domain was visible.

This setup created the foundation for the rest of the lab, including user management, security groups, Group Policy configuration, domain client joining, and authentication testing.

Conclusion

This part of the lab successfully configured the Windows Server virtual machine as the domain controller for the adlab.local domain. The required roles were installed, the domain was created, and Active Directory was verified using the graphical management tools. This prepared the lab environment for creating users, groups, organisational units, and domain-based policies in the next stage.